

Team name:

THE X

Institute name : **RV University**

Theme : **Healthcare**

City:

**Bengaluru
Karnataka**



Problem Statement Title : **Bridging the Healthcare Authorization Gap**

THE PROBLEM

Insured. Yet Uncertain.

A patient with valid health insurance is admitted to a hospital. By discharge, they face unexpected bills, hours of waiting, and zero clarity on what their policy actually covered.



PATIENT

6 in 10 insured patients wait **6 to 48 hours** on discharge day with valid coverage.



HOSPITAL

Insurance desks spend **4 to 6 hours** per case chasing documents, emails, and re-submissions.



INSURER / TPA

Incomplete submissions trigger repeated queries, breaching IRDAI's 1-hour pre-authorization mandate.



Root Cause:

No intelligence layer connecting clinical care, insurance policy, and patient financial liability.

OUR SOLUTION

Meet MedPass AI

The Intelligence Layer for Healthcare Authorization.

MedPass AI connects fragmented clinical information with insurance requirements giving hospitals, patients, and insurers clarity on coverage, authorization status, and financial responsibility.



Authorization Readiness

Know if the case is ready before submission.



Coverage Clarity

Know what insurance covers vs. what the patient owes.



Discharge Intelligence

Know what is blocking the patient from going home.



Outcome Learning

Every case makes the next case faster and smarter.



From fragmented records to a submission ready case in minutes, not hours.



Technologies and Methodology



Python /
FastAPI



PostgreSQL
Redis

spaCy
NLP



AWS Textract



LLM + RAG
Architecture



React.js /
TypeScript



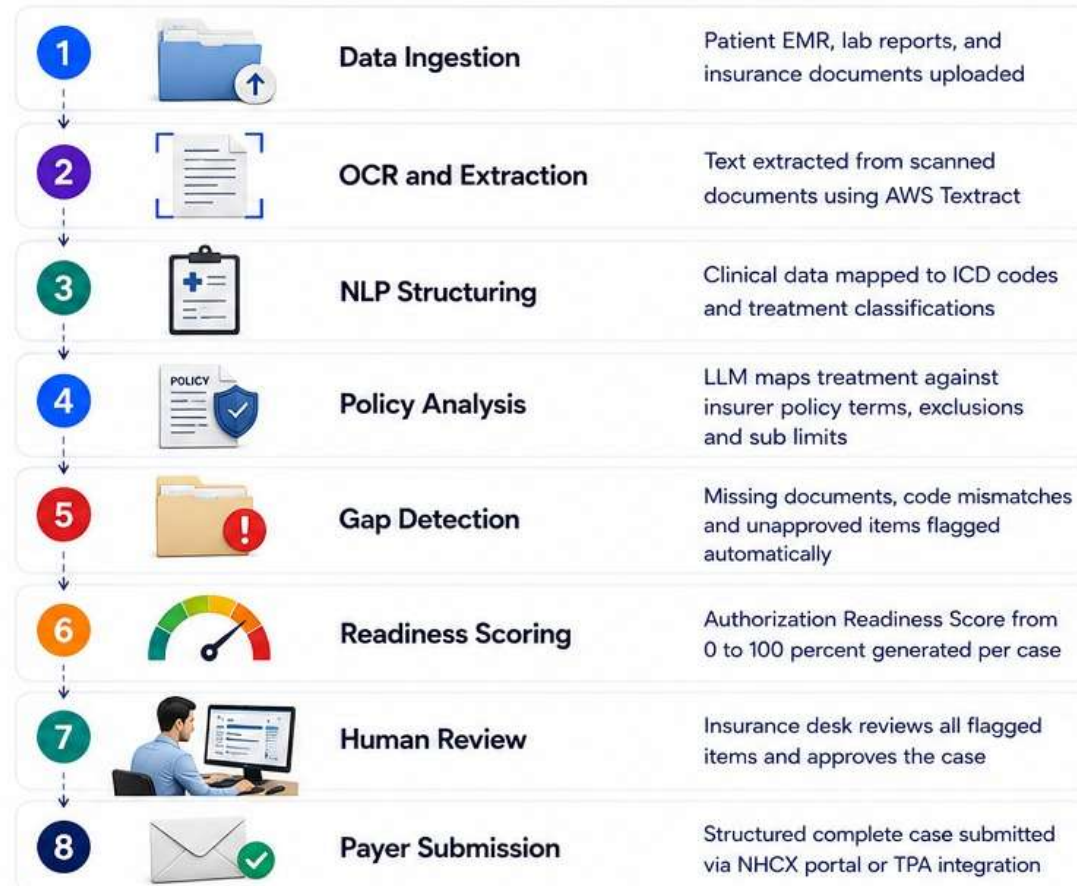
AWS Cloud
HIPAA Ready



NHCX /
TPA API

Implementation Methodology

End to end flow from hospital data to payer submission



System Architecture

Data flow across integrated layers



Higher First Pass Acceptance

Complete and accurate submissions reduce query cycles.



Faster Turnaround

Minutes instead of hours for authorization readiness.



Regulatory Aligned

Supports IRDAI one hour pre-auth mandate and compliance.



Continuous Improvement

Outcome learning improves accuracy with every case.

Is MedPass AI Buildable? Is it Worth Building?

Implementation Roadmap

Four key stages from foundation to intelligence.



Rule Based Engine

Build deterministic validations and rule flows for gap detection.



NLP + OCR Integration

Extract and structure clinical data. Compute authorization readiness.



LLM + NHCX Integration

AI powered policy analysis with RAG. Seamless exchange via NHCX API.



Outcome Learning

Learn from outcomes and feedback to continuously improve accuracy and insights.

Potential Challenges and Risks



DATA PRIVACY

Patient health records are sensitive. DPDP Act 2023 and IRDAI consent requirements mandate strict data governance and explicit patient authorization.

HIGH



LEGACY INTEGRATION

Many hospitals use paper based or offline systems. A phased API integration approach is required to minimize workflow disruption.

MEDIUM



POLICY VARIATION

50 plus insurers operate with different coverage rules, sub-limits, and exclusions. The policy knowledge base requires continuous maintenance.

MEDIUM



REGULATORY SHIFTS

IRDAI rules continue to evolve. The product architecture must be modular enough to adapt to regulatory updates without full redevelopment.

LOW

Potential Impact and Benefits

6 to 48h
Under 3h

Discharge Wait
IRDAI Target Met

50%

Reduction in Insurer
Queries Per Case

500M+

Insured Lives in India
Addressable Market



Social

Patients discharged faster with less financial shock and greater coverage transparency throughout their hospital stay.



Economic

Hospitals recover revenue faster. Insurers receive cleaner submissions. Administrative costs reduced across all stakeholders.



Operational

Paperless workflows, digitized documentation, and NHCX aligned data exchange replacing manual email and fax processes.

How MedPass AI Creates and Captures Value

KEY PARTNERS



- Hospital chains (multi-speciality, private)
- TPAs : MediAssist, MDIndia, Raksha Health
- Health insurers : Star Health, HDFC Ergo, Niva Bupa
- NHCX / National Health Authority
- EMR vendors : Practo, Insta HMS

KEY ACTIVITIES



- AI model development and training
- Policy rule engine maintenance
- Hospital onboarding and EMR integration
- Regulatory compliance, IRDAI and DPDP
- Pilot management and KPI reporting



KEY RESOURCES

- AI and ML engineering team
- Insurance policy knowledge base
- Secure cloud infrastructure, HIPAA ready
- Hospital integration APIs
- Proprietary claims outcome dataset

VALUE PROPOSITIONS



- Authorization Readiness Score before every submission
- Coverage Clarity: insurance vs patient liability at admission
- Discharge Intelligence : what is blocking clearance
- 50 percent fewer insurer queries per case
- IRDAI-compliant : 1h pre-auth and 3h discharge support
- Outcome Learning : every case improves the next

CUSTOMER RELATIONSHIPS



- Dedicated pilot onboarding support
- Real-time SaaS dashboard access
- Monthly KPI reports : TAT, query rate, bed-days
- Insurance desk staff training
- Continuous product improvement from feedback

CHANNELS



Direct B2B sales to hospital chains



TPA partnership programs



NHCX integration marketplace



Industry events: FICCI Health, CII Health Summit

CUSTOMER SEGMENTS



PRIMARY

Private hospital insurance and RCM teams, 500+ bed hospitals



SECONDARY

TPAs managing high-volume cashless claims



TERTIARY

Health insurers seeking cleaner submissions



COST STRUCTURE



- AI and ML engineering salaries
- Cloud infrastructure, AWS and Azure
- Sales and business development
- Regulatory compliance and legal fees
- Pilot program and onboarding costs



REVENUE STREAMS



SaaS subscription per hospital, tiered by bed count



Per-case processing fee for high volume



TPA licensing model: MedPass embedded in TPA workflow



Premium analytics and outcome intelligence dashboard